

Chapter 8

Simple Harmonic Oscillator

Learning goals:

- (1) Describe how classical equations of motion are employed to determine the quantum harmonic oscillator Hamiltonian.
- (2) Derive the states of definite energy for the harmonic oscillator using the intertwining relation.
- (3) Explain how we determine states of definite position and momentum.
- (4) Describe how the uncertainty principle implies that classical and quantum descriptions must be different.
- (5) Construct the wavefunctions for definite-energy states of the harmonic oscillator.

Material to review:

- (a) The complex exponential function and its derivative.
- (b) Arithmetic of complex numbers and their complex conjugate.
- (c) The solution for the classical motion of a harmonic oscillator.
- (d) The canonical commutation relation.

The harmonic oscillator is one of the most important systems in all of physics. This is because many potentials look like harmonic oscillators near their minimum and because the harmonic oscillator is an exactly solvable potential. In this chapter, we show how the classical harmonic oscillator relates to the quantum case and we also discuss many of the properties of the harmonic oscillator solutions.

8.1 Solving the classical harmonic oscillator

We start with the classical harmonic oscillator and show how to solve it. But our solution is not the standard one you saw in your introductory class or that we

discussed in Chap. 1. No, this approach was highlighted by Born and Jordan in their 1930 textbook entitled *Elementare Quantenmechanik* [1]. They showed an interesting approach, which starts from the first-order equations of motion. This is something you might not be familiar with, yet it is something that you should be able to follow, because we are not introducing any concepts you have not seen before. It is just something you will find unfamiliar to you. This means it will take some time for it to feel familiar.

If the harmonic oscillator has mass m and frequency ω , then the two first-order equations of motion are

$$\dot{x} = \frac{p}{m} \quad \text{and} \quad \dot{p} = -m\omega^2 x. \quad (8.1)$$

The traditional way to solve this is to take an extra derivative of the first equation with respect to time, and then use the second equation to obtain a differential equation for position only. But, we are not going to do that here.

Instead, we take a different approach, which involves decoupling the equations of motion. As you can see a derivative of position gives momentum and a derivative of momentum gives position—this is what is meant by saying the equations are coupled. To decouple them, we want to find a linear combination of x and p such that the derivative is proportional to that very same linear combination. This will then lead to a simple differential equation that we can immediately solve. Let us give it a try now.

We first form the linear combination

$$A = \frac{1}{\sqrt{2m}}(p + \alpha x), \quad (8.2)$$

where α is not yet known, and then take the time derivative to obtain

$$\dot{A} = \frac{1}{\sqrt{2m}}(\dot{p} + \alpha \dot{x}) = \frac{1}{\sqrt{2m}}\left(-m\omega^2 x + \frac{\alpha p}{m}\right). \quad (8.3)$$

(Note that we included a factor of $1/\sqrt{2m}$ because it makes a number of formulas simpler later.) Next, we factor the coefficient of the p term out. This yields

$$\dot{A} = \frac{\alpha}{m} \frac{1}{\sqrt{2m}} \left(p - \frac{m^2 \omega^2}{\alpha} x \right). \quad (8.4)$$

We want this to be equal to a constant times A . This means, we must have

$$\alpha = -\frac{m^2 \omega^2}{\alpha}, \quad (8.5)$$

which is solved by $\alpha = \pm im\omega$. Notice that this is a pure imaginary number. Let us take the $-$ solution and denote that as A . Then, we have

$$\dot{A} = -i\omega A. \quad (8.6)$$

One can see that A is a function whose derivative is proportional to A . We learned what function this is early on in calculus. The solution is a complex exponential function, with a multiplicative constant. Hence, we have

$$A(t) = A_0 e^{-i\omega t} = \frac{1}{\sqrt{2m}}(p(t) - im\omega x(t)). \quad (8.7)$$

Here, $A_0 = (p_0 - im\omega x_0)/\sqrt{2m}$, which is found by setting $t = 0$, with x_0 the initial position and p_0 the initial momentum. If we choose the positive sign, the solution turns out to be the complex conjugate of S , so we have

$$A^*(t) = A_0^* e^{i\omega t} = \frac{1}{\sqrt{2m}}(p(t) + im\omega x(t)). \quad (8.8)$$

The $*$ denotes the complex conjugate.

We have decoupled the differential equations, solved them, and found two solutions. How do we use these solutions? Well, we want to find $x(t)$ and $p(t)$. How can we do this? If we look at A and A^* , we see we can take linear combinations to isolate x and p . So, we have that

$$x(t) = \frac{\sqrt{2m}(A(t) - A^*(t))}{-2im\omega} \text{ and } p(t) = \frac{\sqrt{2m}(A(t) + A^*(t))}{2}. \quad (8.9)$$

Simplifying the algebra, we find

$$x(t) = \frac{\sqrt{2m}(A_0 e^{-i\omega t} - A_0^* e^{i\omega t})}{-2im\omega} = x_0 \cos(\omega t) + \frac{p_0}{m\omega} \sin(\omega t), \quad (8.10)$$

which is a result you should work out on your own. Similarly, we have

$$p(t) = \frac{\sqrt{2m}(A_0 e^{-i\omega t} + A_0^* e^{i\omega t})}{2} = p_0 \cos(\omega t) - m\omega x_0 \sin(\omega t), \quad (8.11)$$

which is the conventional solution for the harmonic oscillator. Note that both of these derivations used the definitions of cosine and sine, given by $\cos(\omega t) = \frac{1}{2}(e^{i\omega t} + e^{-i\omega t})$ and $\sin(\omega t) = \frac{1}{2i}(e^{i\omega t} - e^{-i\omega t})$.

I hope you agree this is a neat way to solve for the motion of the simple harmonic oscillator. But, there is more. Let us look at the product of the two functions:

$$A^*(t)A(t) = \underbrace{\frac{p^2(t)}{2m} + \frac{1}{2}m\omega^2 x^2(t)}_{\text{Energy}} = |A_0|^2 e^{i\omega t} e^{-i\omega t} = \underbrace{\frac{p_0^2}{2m} + \frac{1}{2}m\omega^2 x_0^2}_{\text{constant}}, \quad (8.12)$$

which is a simple way to show that energy is conserved.

Now, we went through this material a bit quickly. Please take the time to re-read through it and be sure you follow it, as you need to have a strong foundation in this before we jump to the quantum case.

You might be asking why solve it this way? It was interesting, but why do we do this? The answer comes from what we do next, as we will find thinking about the classical solution this way leads to a similar quantum treatment. As we have done before, our approach is to try to relate quantum to classical as much as we can.

8.2 Classical to quantum correspondence

In addition to following the Born and Jordan textbook, this section also follows a more modernized treatment [7]. We want to relate the classical mechanics we just developed to the quantum world. In quantum mechanics, one of the most important states we want to find are states that have definite energy. So, we need to find the energy operator (which is called the Hamiltonian) and employ it to determine states of definite energy. What hints do we have from our classical analysis? Well, we saw that the energy is given by $A^*(t)A(t)$, so we try to generalize this to quantum mechanics. To do this we must replace $x \rightarrow \hat{x}$ and $p \rightarrow \hat{p}$ in the new operators $\hat{A}^\dagger$ and $\hat{A}$, and to find the Hamiltonian, it is suggested to look at $\hat{A}^\dagger \hat{A}$; note that the quantum Hamiltonian takes the same form as the classical one except we use operators, so $\hat{H} = \frac{1}{2m}\hat{p}^2 + \frac{1}{2}m\omega^2\hat{x}^2$. In doing this, we need to recall the canonical commutation relation, where $[\hat{x}, \hat{p}] = i\hbar$.

So, let us examine this product. We find that

$$\begin{aligned}\hat{A}^\dagger \hat{A} &= \frac{1}{2m}(\hat{p} + im\omega\hat{x})(\hat{p} - im\omega\hat{x}) \\ &= \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 - \frac{i}{2}\omega[\hat{p}, \hat{x}] \\ &= \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 - \frac{1}{2}\hbar\omega\end{aligned}\tag{8.13}$$

or

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 = \hat{A}^\dagger \hat{A} + \frac{1}{2}\hbar\omega.\tag{8.14}$$

The quantum Hamiltonian has an extra constant in it when compared to the classical energy.

How do we work with the quantum Hamiltonian to find states with definite energy? This is the main task of the next section. Here, we will focus on the lowest-energy state and explain how to determine it. It is called the ground state. In the next section, we describe how to find the other states with definite energy.

To begin, we recall that we denote abstract states as kets using the Dirac notation. The ground state is usually denoted with a 0, so the ground state will be the state $|0\rangle$. It will have the lowest possible energy of all definite energy states. Note that because the harmonic oscillator potential is a positive semidefinite function (meaning it is larger or equal to zero), in classical mechanics, we know that the energy of the

harmonic oscillator is larger than or equal to zero. The same will hold in quantum mechanics—indeed we will find it is always larger than zero here.

Before we start, we need to review what the norm of a vector is. If we had a 3-tuple, such as $\vec{v} = (c_1, c_2, c_3)$ with all of the components complex numbers, then we know the norm is the inner product of $\vec{v}$ with itself, except we need to take the complex conjugate of the row vector. Hence, $\|\vec{v}\|^2 = |c_1|^2 + |c_2|^2 + |c_3|^2$. This obviously generalizes to higher dimensions in a simple way. Now, here comes the important property—if the norm is zero, then $c_i = 0$ for all i and similarly if $c_i = 0$ for all i , then the norm is zero. There is only one vector with zero norm, and it is the zero vector $(0, 0, 0)$.

Okay, back to the harmonic oscillator. Let us suppose that $|0\rangle$ is the ground state, such that $\hat{H}|0\rangle = E_0|0\rangle$. Then, if we assume the ground state is normalized, so $\langle 0|0\rangle = 1$, we find

$$E_0 = \langle 0|\hat{H}|0\rangle. \quad (8.15)$$

Be sure you follow the logic for this.

Our next step is to evaluate the expectation value using our form for $\hat{H}$. It has two terms:

$$E_0 = \langle 0|\hat{A}^\dagger \hat{A}|0\rangle + \langle 0|\frac{\hbar\omega}{2}|0\rangle = \|\hat{A}|0\rangle\|^2 + \frac{\hbar\omega}{2}. \quad (8.16)$$

We obtain the norm, because $(\hat{A}|0\rangle)^\dagger = \langle 0|\hat{A}^\dagger$ via our rules, and we can break the factorized form of the Hamiltonian into the norm of a single vector plus a constant. Now, we immediately know what the minimum is—it is the state where the vector is the zero vector, so we have

$$\hat{A}|0\rangle = 0, \quad (8.17)$$

which is a condition called the subsidiary condition, and an important one to use in solving problems abstractly with operators. Hence, we have found the ground state energy. Note, it is not zero. The fact that it is nonzero is called the zero-point energy of the system.

Because the expectation value of the Hamiltonian maps to a norm squared plus a constant, we say that the Hamiltonian operator is written in a positive semidefinite form (because we do not worry about the overall constant term).

This is our first step towards understanding the simple harmonic oscillator. Recall what we did—we took the form of the classical energy to motivate how we construct the quantum Hamiltonian, and then we used some insight about the properties of norms to determine an equation the ground state must satisfy and to find the ground-state energy for the quantum oscillator.

This is pretty cool stuff!

8.3 States of definite energy

Our next job is to determine the other states with definite energy. We will construct them from the ground state plus operators acting on the ground state. The procedure

is likely to be quite unfamiliar to you. So, please take the time to go through it carefully. The approach we discuss here was first worked out in this form by Erwin Schrödinger in 1940 [6]. An image of Schrödinger from about this time is in Fig. 8.1.



Fig. 8.1: Erwin Schrödinger in the late 1930s, around the time he invented the factorization method. Schrödinger was an important figure in the development of quantum mechanics, perhaps best known for the Schrödinger equation. But, he also was a sexual predator. He is someone who challenges us to recognize the fantastic science, but to temper that recognition given his extreme failings as a person. [Image credit: AIP Emilio Segrè Visual Archives, Physics Today Collection. Converted to a stencil by the authors.]

Our first step is to derive the intertwining relation. This asks the question, what happens if I want to rearrange the order of $\hat{H}\hat{A}^\dagger$ to have the $\hat{A}^\dagger$ appear in the left hand side? This is nontrivial, because both the Hamiltonian and $\hat{A}^\dagger$ are operators, and we know operators often do not commute.

To derive the intertwining relation, we just use our definitions. We do have to use the commutation relation $[\hat{A}, \hat{A}^\dagger] = \hbar\omega$ as well (which you should work out for yourself). So, we have

$$\begin{aligned}
 \hat{H}\hat{A}^\dagger &= \left(\hat{A}^\dagger\hat{A} + \frac{\hbar\omega}{2}\right)\hat{A}^\dagger = \left(\hat{A}^\dagger\hat{A}\hat{A}^\dagger + \frac{\hbar\omega}{2}\hat{A}^\dagger\right) = \hat{A}^\dagger\left(\hat{A}\hat{A}^\dagger + \frac{\hbar\omega}{2}\right) \\
 &= \hat{A}^\dagger\left(\hat{A}^\dagger\hat{A} + [\hat{A}, \hat{A}^\dagger] + \frac{\hbar\omega}{2}\right) = \hat{A}^\dagger\left(\hat{A}^\dagger\hat{A} + \frac{3\hbar\omega}{2}\right) \\
 &= \hat{A}^\dagger(\hat{H} + \hbar\omega).
 \end{aligned} \tag{8.18}$$

You can see that as we move the $\hat{A}^\dagger$ in from the right and then out from the left we then need to use a commutator to re-arrange it into the Hamiltonian plus something. In words, we can move a Hamiltonian past an $\hat{A}^\dagger$ at the expense of shifting the Hamiltonian by $\hbar\omega$. This then allows us to find the energy eigenstates, although it might not seem like it at the moment. So, let us show that next.

The claim is that the state $(\hat{A}^\dagger)^n |0\rangle$ is an (unnormalized) energy eigenstate with energy $E_n = \hbar\omega(n + \frac{1}{2})$. To check this, we use the intertwining relation

$$\hat{H}(\hat{A}^\dagger)^n |0\rangle = \hat{A}^\dagger(\hat{H} + \hbar\omega)(\hat{A}^\dagger)^{n-1} |0\rangle \cdots = (\hat{A}^\dagger)^n (\hat{H} + \hbar\omega n) |0\rangle = E_n (\hat{A}^\dagger)^n |0\rangle. \quad (8.19)$$

Since $\hat{H}$ acting on the state returns a number times that state, it is an eigenvector-eigenvalue relationship. It turns out these are all of the energy eigenstates, although that might not be obvious at this point.

Now, we compute the normalization factor. We have $|n\rangle = C_n (\hat{A}^\dagger)^n |0\rangle$. Then

$$1 = |C_n|^2 \langle 0 | \hat{A}^n (\hat{A}^\dagger)^n | 0 \rangle \quad (8.20)$$

and we need to find C_n . To do this, we peel the rightmost $\hat{A}$ and the leftmost $\hat{A}^\dagger$ and note that $\hat{A}\hat{A}^\dagger = \hat{H} + \frac{\hbar\omega}{2}$. After passing it through $n-1$ $\hat{A}^\dagger$ s, we find the operator $\hat{H} + \hbar\omega(n - \frac{1}{2})$ acting on $|0\rangle$. Recalling the ground state-energy, this becomes $\hbar\omega n$. We can then repeat the same approach $n-1$ more times, and we find

$$C_n = \frac{1}{\sqrt{(\hbar\omega)^n n!}}, \quad (8.21)$$

where we chose to make the normalization constant a positive real number. Hence, we have

$$\hat{H} |n\rangle = E_n |n\rangle \quad \text{with} \quad E_n = \hbar\omega(n + \frac{1}{2}) \quad \text{and} \quad |n\rangle = \frac{1}{\sqrt{(\hbar\omega)^n n!}} (\hat{A}^\dagger)^n |0\rangle. \quad (8.22)$$

That is all it takes to find the definite energy states of the simple harmonic oscillator!

One can see that the definite energies in these different states are discrete. What does this mean, and why did it happen? The second question is the easier one to answer. If we look carefully at its origin, we see it can be traced to the discrete shift in the Hamiltonian from the intertwining relation, which occurred because the canonical commutation relation is nonzero. That is all there is to it! One cannot escape energies increasing by discrete amounts for bound states (continuum states do have continuously varying energies, but there are no continuum states in the harmonic oscillator). The first question is much more subtle and often is not understood properly by students, so we will discuss it carefully.

The key point is if you want a state with definite energy, then the energy takes on one of these special discrete values. But, if we allow for superpositions of different energy states, we see that even with superpositions of bound states, we can have

continuously varying average energies. This is discussed further in Prob. 8.2 with a concrete example. We will see another clear example of this when we discuss the coherent states of the simple harmonic oscillator.

We also found that the ground-state energy is nonzero. This energy is called the zero-point energy. One cannot have an equilibrium quantum state with an energy below the zero-point energy. This is a point that also can be confused. But, there are situations where one can remove energy from the ground state, if it is strongly coupled to another system and the ground-state energy of the new coupled system lies below the initial ground-state energy. Sounds subtle, and it is. One needs to always be careful about absolute statements in science.

We end the section with a very special result—there are operators that carry us from one energy eigenstate to its nearest neighbor with energy just above or just below. For example, one can immediately see that $\hat{A}^\dagger |n\rangle = \sqrt{\hbar\omega(n+1)} |n+1\rangle$ and similarly $\hat{A} |n\rangle = \sqrt{\hbar\omega n} |n-1\rangle$ (see if you can verify both of these results). What is important to know is that there are very few other systems for which energy raising and lowering operators can be found and it is difficult for those other systems to establish that there are operators that can raise or lower the energy.

8.4 Heisenberg's original solution

In Chapter 7, we developed matrix mechanics with two assumptions—states are described by matrices with the Heisenberg time dependence, and the matrices satisfy the classical equations of motion. Using that, we found that the canonical commutation relation was independent of time, and was proportional to a unit matrix, so

$$\sum_{l=0}^{\infty} (\mathbf{X}_{ml}\mathbf{P}_{ln} - \mathbf{P}_{ml}\mathbf{X}_{ln}) = i\hbar\delta_{mn}, \quad (8.23)$$

where δ_{mn} is the Kronecker delta function, which is 1 if $m = n$ and is 0 otherwise.

Note that the summation for the matrix multiplication started from 0 and ran to infinity. This is because we have a lowest-energy state, and when we work with matrix mechanics in its conventional form, the matrices are organized so that $E_m < E_n$ for $m < n$. Note that this requires there to be no degeneracies, where at least two energies would be equal. This always holds for one-dimensional problems on the full one-dimensional domain.

The Hamiltonian, written in matrix form is

$$\mathbf{H} = \frac{1}{2m}\mathbf{P}^2 + \frac{1}{2}m\omega^2\mathbf{X}^2, \quad (8.24)$$

and is assumed to be a diagonal matrix with energies running upward from 0 to ∞ in increasing order.

We now need to carefully parse the equations of motion. We start as we did at the beginning of the chapter, where the equation of motion are translated into

$$\dot{\mathbf{X}} = \frac{1}{m}\mathbf{P} \leftrightarrow i\omega_{mn}\mathbf{X}_{mn} = \frac{1}{m}\mathbf{P}_{mn} \text{ and } \dot{\mathbf{P}} = -m\omega^2\mathbf{X} \leftrightarrow i\omega_{mn}\mathbf{P}_{mn} = -m\omega^2\mathbf{X}_{mn}. \quad (8.25)$$

Here, $\omega_{mn} = \frac{E_m - E_n}{\hbar}$, for the matrix case. Note that the m that is not a matrix index is the mass. Please do not get confused!

Now, to solve these equations of motion, we take the second relation on the left and substitute it in for the momentum matrix in the second relation on the right and simplify. We find that

$$(\omega_{mn}^2 - \omega^2)\mathbf{X}_{mn} = 0. \quad (8.26)$$

This is all we need to solve the problem. But, it still takes a few more steps. So, here we go.

First notice that $\omega_{mn}^2 - \omega^2 = (\omega_{mn} + \omega)(\omega_{mn} - \omega)$, so that $(\omega_{mn} + \omega)(\omega_{mn} - \omega)\mathbf{X}_{mn} = 0$. This means we have one of three possibilities: either $\omega_{mn} = \omega$, or $\omega_{mn} = -\omega$, or $\mathbf{X}_{mn} = 0$. Which further says that $E_m = E_n \pm \hbar\omega$, for any matrix elements where $\mathbf{X}_{mn} \neq 0$. Second, we examine the canonical commutation relation, which tells us that

$$\sum_{l=0}^{\infty} (\mathbf{X}_{ml}\mathbf{P}_{ln} - \mathbf{P}_{ml}\mathbf{X}_{ln}) = m \sum_{l=0}^{\infty} (i\omega_{ln}\mathbf{X}_{ml}\mathbf{X}_{ln} - i\omega_{ml}\mathbf{X}_{ml}\mathbf{X}_{ln}) = i\hbar\delta_{mn}. \quad (8.27)$$

If we set $m = n$, we find that

$$\sum_{l=0}^{\infty} 2\omega_{ml}|\mathbf{X}_{ml}|^2 = -\frac{\hbar}{m} \quad (8.28)$$

Now, because the energies are nondegenerate, we have that in each row, there are at most only two matrix elements for $\mathbf{X}$ that are nonzero. But, for the $m = 0$ case, there is only one, so, we find that

$$|\mathbf{X}_{01}|^2 = \frac{\hbar}{2m\omega}, \text{ or } \mathbf{X}_{01} = \sqrt{\frac{\hbar}{2m\omega}}e^{-i\omega t}. \quad (8.29)$$

Using our equation of motion, then tells us that $\mathbf{P}_{01} = -im\omega\mathbf{X}_{01} = i\sqrt{\frac{\hbar m\omega}{2}}e^{-i\omega t}$. Finally, we note that the matrix elements are all Hermitian, which means $\mathbf{M}_{mn} = \mathbf{M}_{nm}^*$. This is another assumption, but it is very reasonable when thinking in terms of reciprocity. We already have the exponential time factors are related by the interchange of $m \leftrightarrow n$ and the complex conjugate. If the constant part of the matrix element is real, then it makes sense that the transition from m to n has the same numerical value as the transition from n to m . It only is hard to argue if the matrix element is not real—but for this problem they are real—so it is not an issue, and the Hermiticity assumption turns out to be the correct assumption for the general case.

Okay, we now put this together to find the 00 matrix element of $\mathbf{H}$. We have

$$\mathbf{H}_{00} = \frac{1}{2m}|\mathbf{P}_{01}|^2 + \frac{1}{2}m\omega^2|\mathbf{X}_{01}|^2 = \frac{\hbar\omega}{2}. \quad (8.30)$$

Of course, this is the same as what we found earlier in the chapter. Using the fact that the energy increases by $\hbar\omega$ for each step, we then find that $E_n = \hbar\omega(n + \frac{1}{2})$.

We have found all of the energies (and hence, the $\mathbf{H}$ matrix). What about the position and momentum matrices? Let us first focus on position. We have already found that $X_{10} = \sqrt{\hbar/2m\omega}e^{i\omega t}$. Using the recurrence relation in Eq. (8.28), we have

$$2\omega|\mathbf{X}_{10}|^2 - 2\omega|\mathbf{X}_{12}|^2 = -\frac{\hbar}{m}, \text{ or } |\mathbf{X}_{12}|^2 = \frac{\hbar}{m}, \text{ or } \mathbf{X}_{12} = \sqrt{2}\sqrt{\frac{\hbar}{2m\omega}}e^{-i\omega t}. \quad (8.31)$$

Proceeding similarly, we find $\mathbf{X}_{nn+1} = \sqrt{n+1}\sqrt{\frac{\hbar}{2m\omega}}e^{-i\omega t}$. Hence, the $\mathbf{X}$ matrix looks like

$$\mathbf{X} = \sqrt{\frac{\hbar}{2m\omega}} \begin{pmatrix} 0 & e^{-i\omega t} & 0 & 0 & \dots & 0 \\ e^{i\omega t} & 0 & \sqrt{2}e^{-i\omega t} & 0 & \dots & 0 \\ 0 & \sqrt{2}e^{i\omega t} & 0 & \sqrt{3}e^{-i\omega t} & \dots & 0 \\ 0 & 0 & \sqrt{3}e^{i\omega t} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \dots & 0 \end{pmatrix}, \quad (8.32)$$

which is an infinite sized matrix. In Prob. 8.3, you will compute the $\mathbf{P}$ matrix, and show how to compute the $\mathbf{H}$ matrix and the canonical commutation relation from both of them.

This approach to get the solution did not require too many steps, although it required significant thought to find the pathway needed to generate the final results. Can it be applied to other systems? The Heisenberg, Born, and Jordan ideology was to find matrices for position and momentum that satisfied the canonical commutation relation and then somehow try to find a way to use them to find a diagonal Hamiltonian. It just didn't work for any other problem. Hence, the approach lost favor to other methods. Interestingly, there was an approach they could have tried, using something called the factorization method, that would have worked, but required working with state vectors in addition to the matrices. A recent publication describes how this works [3].

8.5 Uncertainty

In this section, we discuss the Heisenberg uncertainty principle [4], within the context of what are called coherent states, first discovered by Erwin Schrödinger in 1926 [5]. Coherent states can be thought of as starting in the lowest energy, or ground state, and “displacing” the state both in position by x_0 and in momentum by p_0 . The operator that displaces both of these together is called the displacement operator.

The displacement operator $\hat{D}(x_0, p_0)$ takes the form

$$\hat{D}(x_0, p_0) = e^{\frac{i}{\hbar}(-x_0 \hat{p} + p_0 \hat{x})}. \quad (8.33)$$

How do we know this operator provides such displacements? We investigate this by starting with the ground state of the harmonic oscillator $|0\rangle$, which satisfies the subsidiary condition $\hat{A}|0\rangle = 0$ and its Hermitian conjugate $\langle 0|\hat{A}^\dagger = 0$. Now, using the facts that

$$\hat{x} = \frac{\hat{A}^\dagger - \hat{A}}{i\sqrt{2m\omega}} \quad \text{and} \quad \hat{p} = \frac{(\hat{A} + \hat{A}^\dagger)\sqrt{m}}{\sqrt{2}}, \quad (8.34)$$

we can see that the expectation value of position and momentum is zero in the ground state:

$$\langle 0|\hat{x}|0\rangle = 0 \quad \text{and} \quad \langle 0|\hat{p}|0\rangle = 0, \quad (8.35)$$

because $\hat{A}$ annihilates $|0\rangle$ when acting to the right and $\hat{A}^\dagger$ annihilates $\langle 0|$ when acting to the left. It is then fair to say that, on average, the ground state is located at the origin and has no net momentum.

Our next task is then to see what happens to the expectation value of the position and momentum when we displace the ground state. But, before we do that, we need to develop an identity, which is called the Weyl form of the Hadamard lemma. Hermann Weyl was a German mathematical physicist who played an instrumental role in putting much of quantum mechanics onto a more rigorous mathematical footing. Hadamard was an excellent French mathematician working in the late 19th and early 20th centuries. You can see them both in Fig. 8.2.

The identity we need to work out is to calculate

$$e^{\hat{A}} \hat{B} e^{-\hat{A}}, \quad \text{when } [\hat{A}, \hat{B}] \text{ commutes with } \hat{A}. \quad (8.36)$$

Now, at first, this seems like a really difficult operator identity to work out. So, the key is to start with what we know. We use the fact that we know the derivative of an exponential is proportional to the exponential. Does the same hold true for exponentials of operators? Well, we can try to work this out directly. Note, we always differentiate with respect to numerical variables, and never with respect to operators. So consider the following set of equations:

$$\begin{aligned} \frac{d}{d\lambda} e^{\lambda \hat{A}} &= \frac{d}{d\lambda} \sum_{n=0}^{\infty} \frac{1}{n!} \lambda^n (\hat{A})^n = \sum_{n=1}^{\infty} \frac{1}{(n-1)!} \lambda^{n-1} (\hat{A})^n \\ &= \hat{A} \sum_{n=0}^{\infty} \frac{1}{n!} \lambda^n (\hat{A})^n = \hat{A} e^{\lambda \hat{A}} = e^{\lambda \hat{A}} \hat{A}. \end{aligned} \quad (8.37)$$

We simply expand the exponential in its series, bring the derivative into the series and differentiate term-by-term, shift the index of the summation, and recognize the exponential function. The last step, which shows the $\hat{A}$ operator can appear to the left or to the right of the exponential holds because the operator $\hat{A}$ commutes with any function of $\hat{A}$, which can be seen by expanding the function in a power series and noting that $\hat{A}(\hat{A})^n = (\hat{A})^{n+1} = (\hat{A})^n \hat{A}$. This is a powerful result. Let us find a way to use it.



Fig. 8.2: Jacques Hadamard (left) and Hermann Weyl in the 1930s (right). Hadamard was a French mathematician and Weyl was a German mathematical physicist. [Hadamard image is public domain, Weyl image is Creative Commons License Attribution-Share Alike 2.0 Germany. Both were converted to stencils by the authors.]

Let us take the derivative of $\hat{f}(\lambda) = e^{\lambda\hat{A}}\hat{B}e^{-\lambda\hat{A}}$. We find

$$\frac{d\hat{f}(\lambda)}{d\lambda} = e^{\lambda\hat{A}}\hat{A}\hat{B}e^{-\lambda\hat{A}} - e^{\lambda\hat{A}}\hat{B}\hat{A}e^{-\lambda\hat{A}} = e^{\lambda\hat{A}}[\hat{A}, \hat{B}]e^{-\lambda\hat{A}} = [\hat{A}, \hat{B}], \quad (8.38)$$

where we used the fact that $e^{\lambda\hat{A}}e^{-\lambda\hat{A}} = 1$ (because $\hat{A}$ commutes with itself, so it is just like the identity with numbers) and the fact that $[\hat{A}, \hat{B}]$ commutes with $\hat{A}$ (and hence with $e^{\lambda\hat{A}}$ as well). This is a simple equation to integrate—we do so from $\lambda = 0$ to $\lambda = 1$ to find

$$\int_0^1 d\lambda \frac{d\hat{f}(\lambda)}{d\lambda} = \hat{f}(1) - \hat{f}(0) = \int_0^1 d\lambda [\hat{A}, \hat{B}] = [\hat{A}, \hat{B}]. \quad (8.39)$$

Now, noting that $\hat{f}(0) = \hat{B}$ and $\hat{f}(1)$ is the operator we are trying to find the identity for, we immediately see that we have just proven that

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] \text{ when } [\hat{A}, \hat{B}] \text{ commutes with } \hat{A}. \quad (8.40)$$

This is called the Weyl form of the Hadamard lemma.

Now, we examine the average position and momentum of the displaced ground state $|x_0, p_0\rangle = \hat{D}(x_0, p_0)|0\rangle$. We use the identity we just worked out, which says

$$\hat{D}^\dagger(x_0, p_0)\hat{x}\hat{D}(x_0, p_0) = e^{-\frac{i}{\hbar}(-x_0\hat{p}+p_0\hat{x})}\hat{x}e^{\frac{i}{\hbar}(-x_0\hat{p}+p_0\hat{x})} = \hat{x} + \frac{ix_0}{\hbar}[\hat{p}, \hat{x}] = \hat{x} + x_0 \quad (8.41)$$

and

$$\hat{D}^\dagger(x_0, p_0)\hat{p}\hat{D}(x_0, p_0) = e^{-\frac{i}{\hbar}(-x_0\hat{p}+p_0\hat{x})}\hat{p}e^{\frac{i}{\hbar}(-x_0\hat{p}+p_0\hat{x})} = \hat{p} - \frac{ip_0}{\hbar}[\hat{x}, \hat{p}] = \hat{p} + p_0. \quad (8.42)$$

So, we can immediately see that $\langle x_0, p_0 | \hat{x} | x_0, p_0 \rangle = x_0$ and $\langle x_0, p_0 | \hat{p} | x_0, p_0 \rangle = p_0$. The displacement operator has shifted the average position from 0 to x_0 and the average momentum from 0 to p_0 . This is why it is called the displacement operator.

Why have we been saying “average position” and “average momentum?” It is because in quantum mechanics neither of these quantities are definite in energy eigenstates with nonzero potentials. This requirement is a consequence of the famous Heisenberg uncertainty principle and we investigate this next. It also follows from the fact that, in general, a quantum state has only one definite quantity, so if it is definite in energy, it cannot be definite in position or momentum too.

In the classical oscillator, the position and momentum of the oscillator are known exactly at any time, so the instantaneous fluctuation of the position and momentum must be zero. But, as we will shortly see in quantum mechanics, this is not so. Let us compute the uncertainty of position in the coherent state $|x_0, p_0\rangle$. It is defined to be

$$(\Delta x)_{|x_0, p_0\rangle}^2 = \langle x_0, p_0 | \hat{x}^2 | x_0, p_0 \rangle - (\langle x_0, p_0 | \hat{x} | x_0, p_0 \rangle)^2, \quad (8.43)$$

we then find that

$$(\Delta x)_{|x_0, p_0\rangle}^2 = \langle 0 | \hat{D}^\dagger(x_0, p_0) \hat{x} \hat{D}(x_0, p_0) \hat{D}^\dagger(x_0, p_0) \hat{x} \hat{D}(x_0, p_0) | 0 \rangle - x_0^2 = \langle 0 | \hat{x}^2 | 0 \rangle, \quad (8.44)$$

because $\hat{D}^\dagger(x_0, p_0)\hat{x}\hat{D}(x_0, p_0) = \hat{x} + x_0$, and $\langle 0 | \hat{x} | 0 \rangle = 0$. Note that we used a multiply-by-one in between the two $\hat{x}$ operators since $\hat{D}^\dagger(x_0, p_0)\hat{D}(x_0, p_0) = 1$. To evaluate the remaining expectation value, we use the $\hat{A}$ operator and the subsidiary condition:

$$\begin{aligned} \langle 0 | \hat{x}^2 | 0 \rangle &= -\frac{1}{2m\omega^2} \langle 0 | \left((\hat{A}^\dagger)^2 - \hat{A}^\dagger \hat{A} - \hat{A} \hat{A}^\dagger + \hat{A}^2 \right) | 0 \rangle \\ &= \frac{1}{2m\omega^2} \langle 0 | [\hat{A}, \hat{A}^\dagger] | 0 \rangle = \frac{\hbar}{2m\omega}. \end{aligned} \quad (8.45)$$

Note that we used an add-zero trick here, since $\langle 0 | \hat{A} \hat{A}^\dagger | 0 \rangle = \langle 0 | (\hat{A} \hat{A}^\dagger - \hat{A}^\dagger \hat{A}) | 0 \rangle$, because $\hat{A} | 0 \rangle = 0$. As we can see, the fluctuations of the position are nonzero in the quantum world!

A similar calculation for momentum yields

$$(\Delta p)_{|x_0, p_0\rangle}^2 = \frac{\hbar m \omega}{2}. \quad (8.46)$$

The momentum also has a variance to it. Note that both the variance in position and momentum are independent of x_0 and p_0 . One can combine these two results

together and see that

$$(\Delta x)_{|x_0, p_0\rangle} (\Delta p)_{|x_0, p_0\rangle} = \frac{\hbar}{2}. \quad (8.47)$$

This result is called the Heisenberg uncertainty principle. It says that one cannot have quantum states of definite position and definite momentum at the same time. This is a result we expect, but here it is explicitly shown by our calculation. Note that we have not proved the result in general, we just examined it for the coherent states. It is true for all quantum states though. Just note that this does not mean that you cannot measure position and momentum at the same time. The uncertainty principle is often conflated in this fashion. It is just a statement about not being able to have definite position and definite momentum in the same quantum state.

One final thing we would like to do is to calculate the energy of the coherent state. This is easy to work out given what we have done, because we have that

$$\begin{aligned} \langle x_0, p_0 | \hat{H} | x_0, p_0 \rangle &= \frac{1}{2m} \left((\Delta p)_{|x_0, p_0\rangle}^2 + p_0^2 \right) + \frac{1}{2} m \omega^2 \left((\Delta x)_{|x_0, p_0\rangle}^2 + x_0^2 \right) \\ &= \frac{p_0^2}{2m} + \frac{1}{2} m \omega^2 x_0^2 + \frac{\hbar \omega}{2}. \end{aligned} \quad (8.48)$$

In this calculation, we used the fact that $\langle 0 | \hat{x}^2 | 0 \rangle = (\Delta x)_{|x_0, p_0\rangle}^2 + (\langle x_0, p_0 | \hat{x} | x_0, p_0 \rangle)^2$, and similarly for momentum. Hence, what we see is the displaced ground state simply raises the quantum energy by the classical energy corresponding to the displacement. This only happens for the harmonic oscillator—other systems are more complicated. You can see, however, that when the displacements are large, so that the quantum corrections become smaller, then they are unimportant and the system begins to look more and more classical.

We also want to take a moment to point out that the energy is continuously raised from the minimum value it has in the ground state—namely $\frac{\hbar \omega}{2}$. Students often can be confused by this, because the energy eigenstates increase discretely. So how can this occur? It does so because when we make superpositions of energy eigenstates, their average energy can increase continuously, and that is precisely what is happening here, although we have not explicitly shown it for the coherent states.

It is neat how examining the uncertainty helps us understand more clearly the relationship between the classical and the quantum solutions. We will see even more of this relationship when we examine the time dependence of these coherent states in the next section.

8.6 Time evolution of quantum states

In our initial classical calculation for the simple harmonic oscillator in Sec. 8.1 we found the time evolution of the classical harmonic oscillator. Now, we will determine it for the quantum case. To start we have to determine how to evolve states forward in time. Of course, we do this with an operator, called the time-evolution operator

$\hat{U}(t, t_0)$ which evolves any quantum state from time t_0 to time t . Hence, we have

$$|\psi(t)\rangle = \hat{U}(t, t_0) |\psi(t_0)\rangle. \quad (8.49)$$

How do we find this operator? We need to use a couple of different facts. First, we use the semigroup rule, which says the time evolution from t_0 to t_1 is the same as the time evolution from t_0 to t_2 followed by the evolution from t_2 to t_1 . In equations, this says

$$\hat{U}(t_1, t_0) = \hat{U}(t_1, t_2) \hat{U}(t_2, t_0). \quad (8.50)$$

We also need $\hat{U}$ to be a unitary operator. This is because the normalization of a quantum state cannot change in time (because the total probability for something to occur is always one). Third, because the Hamiltonian governs the time evolution in classical systems, it should in quantum systems too. Finally, the argument of a function must be dimensionless. So, we have a target for the evolution operator, given by

$$\hat{U}(t, t_0) = e^{-\frac{i}{\hbar} \hat{H}(t-t_0)}. \quad (8.51)$$

You can see that this operator satisfies all of our criteria. The only piece that we could not infer was whether it has a positive or negative sign in the exponent. It turns out the negative sign makes states move in the direction of their momentum, as opposed to opposite to the direction of the momentum. If you were unhappy with these arguments, then you can simply take this form for the evolution operator as a postulate.

We will consider the time evolution of states, as governed by this operator. If we have states with definite energy, $|n\rangle$ with $\hat{H}|n\rangle = E_n|n\rangle$, then, because they are eigenstates, the time evolution can be immediately determined. So, we have

$$\hat{U}(t, t_0) |n\rangle = e^{-\frac{i}{\hbar} \hat{H}(t-t_0)} |n\rangle = e^{-\frac{i}{\hbar} E_n(t-t_0)} |n\rangle. \quad (8.52)$$

Note that when the function of an operator acts on an eigenstate, we can replace the operator by the eigenvalue in the function, which is what we used here. If you are concerned about whether this is the right thing to do, you can expand the exponential in a power series, act the powers of $\hat{H}$ on the eigenstate, and then sum the series. It gives precisely the result we already found.

If we have a superposition of energy eigenstates, so that

$$|\psi(t_0)\rangle = \sum_{n=0}^{\infty} c_n |n\rangle, \quad (8.53)$$

then the time-evolved state is

$$|\psi(t)\rangle = \sum_{n=0}^{\infty} c_n e^{-\frac{i}{\hbar} E_n(t-t_0)} |n\rangle. \quad (8.54)$$

This is a complicated expression, but it shows the general principle about how we can work with expressions that involve operators. If we can expand our state in terms of the eigenstates of the operator, then we can easily determine the action of the operator on the state, in the same form as what we just did.

What we really want to investigate is how the coherent states evolve in time. We have

$$|x_0, p_0; t\rangle = e^{-\frac{i}{\hbar}\hat{H}t} \hat{D}(x_0, p_0) |0\rangle, \quad (8.55)$$

where we set $t_0 = 0$ for simplicity. To evaluate this, we need to examine the Hadamard lemma for the position and momentum operators. But, it is not in the Weyl form, because the commutator of $\hat{H}$ with $\hat{x}$ or $\hat{p}$ does not commute with $\hat{H}$ (please verify this yourself). To work out the full Hadamard lemma, we use our same derivative approach. We have the two operator functions

$$\hat{x}(t) = e^{-\frac{i}{\hbar}\hat{H}t} \hat{x} e^{\frac{i}{\hbar}\hat{H}t} \quad \text{and} \quad \hat{p}(t) = e^{-\frac{i}{\hbar}\hat{H}t} \hat{p} e^{\frac{i}{\hbar}\hat{H}t}, \quad (8.56)$$

(if you know what a Heisenberg picture operator is, these operators are *not* those; if you do not know what that is, do not worry, as it makes no difference to our calculation). Evaluating the derivatives, we have

$$\frac{d\hat{x}(t)}{dt} = e^{-\frac{i}{\hbar}\hat{H}t} \left(-\frac{i}{\hbar} [\hat{H}, \hat{x}] \right) e^{\frac{i}{\hbar}\hat{H}t} = e^{-\frac{i}{\hbar}\hat{H}t} \left(-\frac{i}{2m\hbar} [\hat{p}^2, \hat{x}] \right) e^{\frac{i}{\hbar}\hat{H}t} = -\frac{\hat{p}(t)}{m} \quad (8.57)$$

and

$$\frac{d\hat{p}(t)}{dt} = e^{-\frac{i}{\hbar}\hat{H}t} \left(-\frac{i}{\hbar} [\hat{H}, \hat{p}] \right) e^{\frac{i}{\hbar}\hat{H}t} = e^{-\frac{i}{\hbar}\hat{H}t} \left(-\frac{im\omega^2}{2\hbar} [\hat{x}^2, \hat{p}] \right) e^{\frac{i}{\hbar}\hat{H}t} = m\omega^2 \hat{x}(t). \quad (8.58)$$

Taking a second derivative, we decouple the equations so that

$$\frac{d^2\hat{x}(t)}{dt^2} = -\omega^2 \hat{x}(t) \quad \text{and} \quad \frac{d^2\hat{p}(t)}{dt^2} = -\omega^2 \hat{p}(t), \quad (8.59)$$

with solution

$$\hat{x}(t) = \cos(\omega t) \hat{x} - \frac{1}{m\omega} \sin(\omega t) \hat{p} \quad \text{and} \quad \hat{p}(t) = \cos(\omega t) \hat{p} + m\omega \sin(\omega t) \hat{x}. \quad (8.60)$$

We are almost there. We have to go back to the problem we are trying to solve, which is to determine the result for Eq. (8.55). So, by using a multiply-by-one, we have

$$\begin{aligned}
|x_0, p_0; t\rangle &= e^{-\frac{i}{\hbar}\hat{H}t} e^{\frac{i}{\hbar}(-x_0\hat{p}+p_0\hat{x})} e^{\frac{i}{\hbar}\hat{H}t} e^{-\frac{i}{\hbar}\hat{H}t} |0\rangle \\
&= \exp\left(-\frac{i}{\hbar}\left(-x_0e^{-\frac{i}{\hbar}\hat{H}t}\hat{p}e^{\frac{i}{\hbar}\hat{H}t} + p_0e^{-\frac{i}{\hbar}\hat{H}t}\hat{x}e^{\frac{i}{\hbar}\hat{H}t}\right)\right) e^{-\frac{i}{\hbar}\hat{H}t} |0\rangle \\
&= e^{-\frac{i}{\hbar}(x_0\hat{p}(t)+p_0\hat{x}(t))} e^{-\frac{i}{\hbar}\hat{H}t} |0\rangle \\
&= \exp\left(-\frac{i}{\hbar}\left((p_0\cos(\omega t) - m\omega x_0\sin(\omega t))\hat{x} \right. \right. \\
&\quad \left. \left. -(x_0\cos(\omega t) + \frac{p_0}{m\omega}\sin(\omega t))\hat{p}\right)\right) e^{-\frac{i\omega t}{2\hbar}} |0\rangle. \quad (8.61)
\end{aligned}$$

That was a lot of algebra, but the final result is simple. The time-evolved coherent state has an overall phase of $e^{-i\omega t/2\hbar}$ and is displaced by $x_0 \rightarrow x_0\cos(\omega t) + \frac{p_0}{m\omega}\sin(\omega t)$ and $p_0 \rightarrow p_0\cos(\omega t) - m\omega x_0\sin(\omega t)$. Note that this is exactly how the classical harmonic oscillator evolves in time. Hence, we have

$$\begin{aligned}
\langle x_0, p_0; t | \hat{x} | x_0, p_0; t \rangle &= x_0\cos(\omega t) + \frac{p_0}{m\omega}\sin(\omega t) \quad \text{and} \\
\langle x_0, p_0; t | \hat{p} | x_0, p_0; t \rangle &= p_0\cos(\omega t) - m\omega x_0\sin(\omega t). \quad (8.62)
\end{aligned}$$

This is exactly the same as the equation of motion we found for the classical harmonic oscillator. The *average* position and momentum of the coherent states have the same equation of motion as the classical harmonic oscillator. We only see a difference when we look at the variance in the quantum case, which is time-independent and worked out in the previous section. In Prob. 8.8, we derive this result in a different way.

8.7 Wavefunctions

If we recall the Weyl form of the Hadamard lemma for the displacement operator and consider two special cases, we find

$$\hat{D}^\dagger(x_0, p_0=0)\hat{x}\hat{D}(x_0, p_0=0) = \hat{x} + x_0 \quad \text{and} \quad \hat{D}^\dagger(x_0=0, p_0)\hat{p}\hat{D}(x_0=0, p_0) = \hat{p} + p_0. \quad (8.63)$$

These two operators translate just position and momentum, respectively. We can use them to find the states of definite position and of definite momentum.

To do this, we need to start with a state that is annihilated by the position operator, which we call $|0_x\rangle$, with the x subscript to remind us that the state is annihilated by the $\hat{x}$ operator. This means $\hat{x}|0_x\rangle = 0$, which is an eigenvalue-eigenvector relationship for the position eigenvalue $x = 0$, and hence, it is the eigenstate of position at the origin. Now, we claim that

$$\hat{D}(x_0, p_0=0)|0_x\rangle = e^{-\frac{i}{\hbar}x_0\hat{p}}|0_x\rangle = |x_0\rangle, \quad (8.64)$$

is the position eigenstate with eigenvalue x_0 . Let us check:

$$\begin{aligned}
\hat{x} |x_0\rangle &= \hat{x} e^{-\frac{i}{\hbar} x_0 \hat{p}} |0_x\rangle = \underbrace{e^{-\frac{i}{\hbar} x_0 \hat{p}} e^{\frac{i}{\hbar} x_0 \hat{p}}}_{\text{multiply by one}} \hat{x} e^{-\frac{i}{\hbar} x_0 \hat{p}} |0_x\rangle \\
&= e^{-\frac{i}{\hbar} x_0 \hat{p}} \underbrace{e^{\frac{i}{\hbar} x_0 \hat{p}} \hat{x} e^{-\frac{i}{\hbar} x_0 \hat{p}}}_{\text{Weyl form of Hadamard}} |0_x\rangle \\
&= e^{-\frac{i}{\hbar} x_0 \hat{p}} (\hat{x} + x_0) |0_x\rangle = e^{-\frac{i}{\hbar} x_0 \hat{p}} x_0 |0_x\rangle \\
&= x_0 |x_0\rangle .
\end{aligned} \tag{8.65}$$

In this calculation, we wish to establish that $|x_0\rangle$ is an eigenvector of the position operator with an eigenvalue of x_0 . We first use the definition of the state $|x_0\rangle$ and then use a multiply-by-one to set up for the Weyl form of the Hadamard identity, which then is evaluated. Finally, we use the fact that $\hat{x} |0_x\rangle = 0$ and that the number x_0 commutes with operators, and lastly the definition of $|x_0\rangle$ again. So, we can construct states with definite position rather simply from the position eigenstate at the origin.

The same holds for momentum. Our claim is that is $\hat{p} |0_p\rangle = 0$, then

$$|p_0\rangle = e^{\frac{i}{\hbar} p_0 \hat{x}} |0_p\rangle . \tag{8.66}$$

The proof that $\hat{p} |p_0\rangle = p_0 |p_0\rangle$ is given in Prob. 8.9.

Our next step is to calculate some wavefunctions. Recall, as we said before when we were discussing light in Chap. 4 that the quantum state is the set of rules used to prepare a state from some initial state, while the wavefunction says evaluate those rules at a specific position, or at a specific momentum (if appropriate). In equations, we have that the position-space wavefunction $\psi(x)$ is found from the inner product of the quantum state with a state of definite position, so

$$\psi(x) = \langle x | \psi \rangle . \tag{8.67}$$

The same holds for momentum-space wavefunctions, where $\psi(p) = \langle p | \psi \rangle$. So, can we compute wavefunctions for the states of definite position and momentum? The answer is yes and no. We can compute the momentum-space wavefunction of a state with definite position and we can compute the position-space wavefunction of a state with definite momentum. But the position-space wavefunction of a state of definite position (and similarly for momentum) is a complicated object called a Dirac delta function, and we will defer discussing that object for now.

So, let us find the position-space wavefunction for momentum. It is

$$\psi_p(x) = \langle x | p \rangle = \langle x | e^{\frac{i}{\hbar} p \hat{x}} | 0_p \rangle = e^{\frac{i}{\hbar} x p} \langle x | 0_p \rangle = e^{\frac{i}{\hbar} x p} \langle 0_x | e^{\frac{i}{\hbar} x \hat{p}} | 0_p \rangle = e^{\frac{i}{\hbar} x p} \langle 0_x | 0_p \rangle . \tag{8.68}$$

This is evaluated by first using the definition of $|p\rangle$, evaluating the operator to the left, because it acts on a state with definite position. This becomes a number, which can be removed from the bra-ket. Then we use the definition of $\langle x |$ and operate the operator to the right on a state of definite momentum (at the origin) leaving us with an exponential function multiplied by the number $\langle 0_x | 0_p \rangle$. Determining this number is not so simple, because the state is not normalizable (which occurs for all

states whose eigenvalues change continuously, as opposed to discretely). It can be normalized using what is called delta-function normalization. But, this is a rather technical subject, and it turns out we will not need to know this constant for anything we do now, so we will just carry it along as a constant until we need to determine precisely what it is. You might be interested to know that Weyl is the person who first figured out how to understand the normalization of these states, that do not seem like they can be normalized.

We can similarly work out the momentum-space wavefunction of position, and we find it is given by

$$\psi_x(p) = \langle p|x \rangle = e^{-\frac{i}{\hbar}xp} \langle 0_p|0_x \rangle. \quad (8.69)$$

It is the complex conjugate of the position-space wavefunction of a definite-momentum state.

One important exercise we can perform with these states is to find the matrix representation of the momentum operator in position space. Doing so will aid us significantly in the calculation of the harmonic oscillator wavefunctions. So, we go through this exercise now. You will see that the normalization constant needed in the wavefunction enters and then disappears, without needing to know precisely what it is.

To start, consider that we wish to evaluate the action of momentum on a state $|\psi\rangle$ and we want the wavefunction of the vector that emerges afterwards. Because we do not know what the momentum operator does when it acts on a general state $|\psi\rangle$ or on the position bra $\langle x|$, we need to find a way to introduce states of definite momentum. It turns out this is easy with the multiply-by-one trick because we have what is called a completeness relation, so that

$$\int_{-\infty}^{\infty} dp |p\rangle \langle p| = \hat{\mathbb{I}}, \quad (8.70)$$

which is the identity operator. Hence, we have

$$\begin{aligned} \langle x|\hat{p}|\psi\rangle &= \int_{-\infty}^{\infty} dp \langle x|\hat{p}|p\rangle \langle p|\psi\rangle = \int_{-\infty}^{\infty} dp p \langle x|p\rangle \langle p|\psi\rangle \\ &= \int_{-\infty}^{\infty} dp p e^{\frac{i}{\hbar}xp} \langle 0_x|0_p\rangle \langle p|\psi\rangle = -i\hbar \frac{d}{dx} \int_{-\infty}^{\infty} dp \langle x|p\rangle \langle p|\psi\rangle \\ &= -i\hbar \frac{d}{dx} \langle x|\psi\rangle = -i\hbar \frac{d\psi(x)}{dx}. \end{aligned} \quad (8.71)$$

The calculation goes as follows: First we introduce the complete set of momentum states so that we can operate the momentum operator onto the state and return a number times that state. Then we write out the explicit form of the position-space wavefunction of the state with definite momentum. Next, we see that the factor of p can also be constructed by taking an appropriate derivative with respect to position weighted by some numbers. Next, we see that we can use the completeness again to remove the momentum states, and then we obtain the final result. What this says

is that when a momentum operator acts inside a bra-ket with a state of definite position as the bra, then it can be removed from the bra-ket and replaced by the position-space matrix representation of $\hat{p} \leftrightarrow -i\hbar d/dx$. Note that even though most text books call the derivative an operator, it strictly is not. More precisely it is the matrix representation of the momentum operator in position space. (Technically speaking it is also multiplied by a delta function as well, but because the matrix form is diagonal, that object is usually dropped, and we do so here, allowing us to defer discussing the delta function.) Note further, how the normalization constant entered, but then departed, just as we said it would.

Now, we have all the tools we need to compute the wavefunctions of the harmonic oscillator energy eigenstates. We start from the subsidiary condition, which says $\hat{A}|0\rangle = \frac{1}{\sqrt{2m}}(\hat{p} - im\omega\hat{x})|0\rangle = 0$. If we multiply by $\langle x|$ from the right and denote $\langle x|0\rangle = \psi_0(x)$ for the ground-state wavefunction, then using our new rule for representing the momentum operator, we find that

$$\langle x|(\hat{p} - im\omega\hat{x})|0\rangle = 0 \implies -i\hbar \frac{d}{dx}\psi_0(x) = im\omega x\psi_0(x). \quad (8.72)$$

The differential equation is solved as follows:

$$\begin{aligned} \frac{\frac{d}{dx}\psi_0(x)}{\psi_0(x)} &= -\frac{m\omega}{\hbar}x \\ \int d(\ln \psi_0(x)) &= -\frac{m\omega}{\hbar} \int dx x \\ \ln \psi_0(x) &= -\frac{m\omega}{2\hbar}x^2 + \ln(c) \\ \psi_0(x) &= ce^{-\frac{m\omega}{2\hbar}x^2}, \end{aligned} \quad (8.73)$$

where $c = \langle 0_x|0\rangle$ is the normalization constant, which we determine next. Recall, we need to evaluate

$$\begin{aligned} 1 = \langle 0|0\rangle &= \int_{-\infty}^{\infty} dx \langle 0|x\rangle \langle x|0\rangle = c^2 \int_{-\infty}^{\infty} dx e^{-\frac{m\omega}{\hbar}x^2} \\ &= c^2 \sqrt{\frac{\hbar}{m\omega}} \int_{-\infty}^{\infty} dy e^{-y^2} = c^2 \sqrt{\frac{\pi\hbar}{m\omega}}. \end{aligned} \quad (8.74)$$

This implies

$$c = \langle 0_x|0\rangle = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \quad \text{and} \quad \psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar}x^2}. \quad (8.75)$$

The higher-energy wavefunctions are a bit more complicated to obtain. We go through the process step-by-step so it can be easily followed. We start with the raising operator $\hat{A}^\dagger = (\hat{p} + im\omega\hat{x})/\sqrt{2m}$ and note that it can be rewritten using our Weyl form of the Hadamard lemma as

$$\hat{A}^\dagger = \frac{1}{\sqrt{2m}} e^{\frac{m\omega}{2\hbar} \hat{x}^2} \hat{p} e^{-\frac{m\omega}{2\hbar} \hat{x}^2} = \frac{1}{\sqrt{2m}} \left(\hat{p} + \frac{m\omega}{2\hbar} [\hat{x}^2, \hat{p}] \right) = \frac{1}{\sqrt{2m}} (p + im\omega \hat{x}). \quad (8.76)$$

Substituting this in for the raising operators, we find that

$$|n\rangle = \frac{1}{\sqrt{(\hbar\omega)^n n!}} (\hat{A}^\dagger)^n |0\rangle = \frac{1}{\sqrt{(2\hbar m\omega)^n n!}} e^{\frac{m\omega}{2\hbar} \hat{x}^2} \hat{p}^n e^{-\frac{m\omega}{2\hbar} \hat{x}^2} |0\rangle. \quad (8.77)$$

Now we use the same approach as before to replace the momentum operators by derivatives after multiplying by the position bra. This yields

$$\begin{aligned} \psi_n(x) = \langle x|n\rangle &= \frac{1}{\sqrt{(2\hbar m\omega)^n n!}} e^{\frac{m\omega}{2\hbar} x^2} (-i\hbar)^n \frac{d^n}{dx^n} \left(e^{-\frac{m\omega}{2\hbar} x^2} \psi_0(x) \right) \\ &= \frac{(i)^n}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar} x^2} \underbrace{(-1)^n \sqrt{\left(\frac{\hbar}{m\omega} \right)^n} e^{\frac{m\omega}{\hbar} x^2} \frac{d^n}{dx^n} e^{-\frac{m\omega}{\hbar} x^2}}_{\text{Rodrigues formula}} \\ &= \frac{(i)^n}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar} x^2} \underbrace{(-1)^n e^{y^2} \frac{d^n}{dy^n} e^{-y^2} \Big|_{y=\sqrt{\frac{m\omega}{\hbar}} \hat{x}}}_{\text{Rodrigues formula}} \\ &= \frac{(i)^n}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar} x^2} H_n \left(\sqrt{\frac{m\omega}{\hbar}} \hat{x} \right) \end{aligned} \quad (8.78)$$

In deriving this result, we used the explicit value of the ground state wavefunction and we used a multiply-by-one on the left to obtain the part of the expression that is denoted Rodrigues formula. Be sure to check the result carefully. We explain what the Rodrigues formula is next, as well as what the Hermite polynomial $H_n(y)$ is. Note that the wavefunction we have derived is already normalized, because of how we constructed it. It is common to drop the i^n phase, because it is a global phase and does not change the probabilities.

Olinde Rodrigues (see Fig. 8.3) was a French mathematician in the early and mid 19th century. He is known for discovering a series of formulas for special polynomials, and the one highlighted here is for the so-called Hermite polynomials. The identity says that

$$H_n(y) = (-1)^n e^{y^2} \frac{d^n}{dy^n} e^{-y^2} \quad (8.79)$$

in what is called the physicists normalization. Let us look carefully at this formula. For $n = 0$, we see we obtain $H_0(y) = 1$. For $n = 1$, we have $H_1(y) = 2y$. And for $n = 2$, we have $H_2(y) = 4y^2 - 2$. Please check these yourself as well. What you can see is that as we take derivatives of the Gaussian on the right, we build up a polynomial in y multiplied by the Gaussian. Then the Gaussian on the left cancels the exponential factor and we are left just with the polynomial.



Fig. 8.3: Olinde Rodrigues, who discovered the Rodrigues formulas for special polynomials. He also was the person who first coined the term *avante garde*. [Original image is public domain and converted to look like a stencil by the authors.]

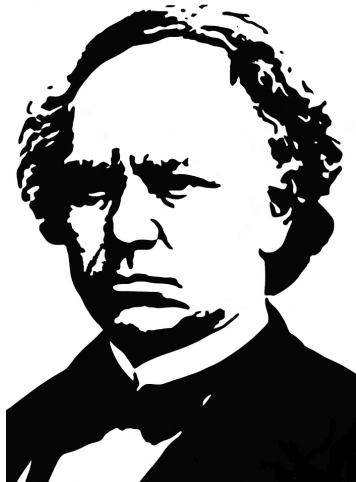


Fig. 8.4: Charles Hermite, who first worked with the Hermite polynomials. [Image is public domain and converted to look like a stencil by the authors.]

How do we determine them in general, because this approach of taking the derivatives is a bit tedious. To sort that out, we need to find the recurrence relations that these polynomials satisfy. This is what the French mathematician Charles Hermite (also mid 19th century) examined; see Fig. 8.4. We will not derive the recurrence relations here, but we will state them. They are

$$H_{n+1}(y) = 2yH_n(y) - 2nH_{n-1}(y). \quad (8.80)$$

The first few Hermite polynomials are summarized in Table 8.1.

n	$H_n(y)$
0	1
1	$2y$
2	$4y^2 - 2$
3	$8y^3 - 12y$
4	$16y^4 - 48y^2 + 12$
5	$32y^5 - 160y^3 + 120y$

Table 8.1: First six Hermite polynomials with the physicist's normalization.

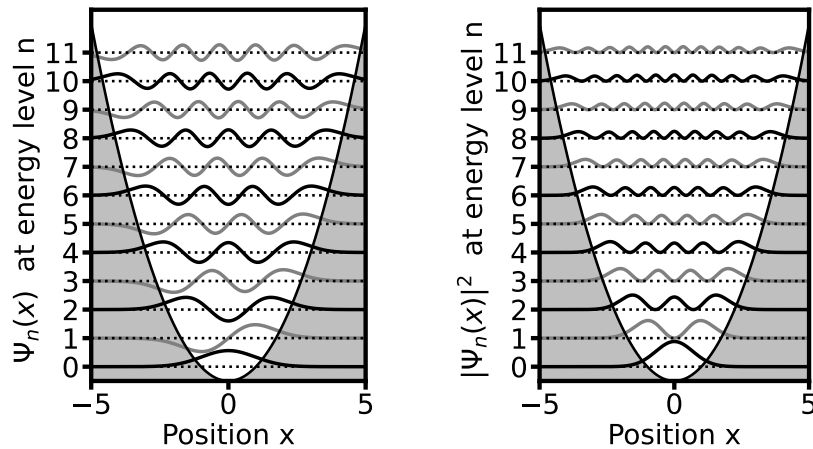


Fig. 8.5: (Left) Wavefunctions of the simple harmonic oscillator and (right) probability distributions of the simple harmonic oscillator. The functions are plotted with the zero located at their respective energy. Note that the functions have been rescaled to be easy to visualize.

In Fig. 8.5, we plot the first 12 wavefunctions $\psi_n(x)$ and probability distribution functions $|\psi_n(x)|^2$. The gray shaded region is the classically forbidden region. You can see that the wavefunctions do leak into the forbidden region, but there is not substantial weight there. As n increases, the wavefunction becomes spread out more uniformly, but it always oscillates and has regions where the probability vanishes.

Can we compare a quantum probability distribution to a classical one? Yes, but only if we consider the classical average over time, which can be converted to a distribution over position. The classical idea is that for a time interval dt , the particle is found in a region of length $dx = v(t)dt$. But what we want is to determine the time interval for a fixed position of size dx . Further, the probability to be in that spatial interval is equal to the ratio of $dt/T_{\frac{1}{2}}$, with $T_{\frac{1}{2}}$ the half period; the half period is used because we traverse the range from $-x_0$ to x_0 only once. Hence, we have

$$P_{\text{cl}}(x) = \frac{dx}{T_{\frac{1}{2}}v(x)} = \frac{\omega dx}{\pi v(x)} = \frac{dx}{\pi \sqrt{x_0^2 - x^2}}, \quad (8.81)$$

where we used the fact that $v(t) = -\omega x_0 \sin(\omega t)$ for $x = x_0 \cos(\omega t)$, which implies that $|v(x)| = \omega x_0 \sin(\omega t) = \omega \sqrt{x_0^2 - x^2}$. At the turning point, where $x = 0$, we see $v \rightarrow 0$. At the origin, we see $|v|$ is maximal. You will investigate the comparison of the classical probability distribution to the quantum one for a large n eigenstate in Prob. 8.14.

We end this section by discussing how to construct the time-evolving wavefunctions of the coherent states. We use a shorthand of $x_0(t) = x_0 \cos(\omega t) + \frac{p_0}{m\omega} \sin(\omega t)$ and $p_0(t) = p_0 \cos(\omega t) - m\omega x_0 \sin(\omega t)$, so that

$$\langle x|x_0, p_0; t \rangle = \langle x|x_0(t), p_0(t) \rangle = \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} \langle x|\hat{D}(x_0(t), p_0(t))|0 \rangle. \quad (8.82)$$

To evaluate this, we need to separate out the position and momentum operators in the displacement operator exponent into separate exponentials. This can be done by using the so-called Weyl form of the Baker-Campbell-Hausdorff formula, which we will derive next. An image of Baker, Campbell, and Hausdorff is shown in Fig. 8.6.

The identity is found by defining $\hat{f}(\lambda) = e^{-\lambda\hat{B}} e^{-\lambda\hat{A}} e^{\lambda(\hat{A}+\hat{B})}$. Then, by taking the derivative and recalling that we are assuming that $[\hat{A}, \hat{B}]$ commutes with $\hat{A}$ and $\hat{B}$, we find that

$$\begin{aligned} \frac{d\hat{f}(\lambda)}{d\lambda} &= -e^{-\lambda\hat{B}} \hat{B} e^{-\lambda\hat{A}} e^{\lambda(\hat{A}+\hat{B})} - e^{-\lambda\hat{B}} e^{-\lambda\hat{A}} \hat{A} e^{\lambda(\hat{A}+\hat{B})} + e^{-\lambda\hat{B}} e^{-\lambda\hat{A}} (\hat{A} + \hat{B}) e^{\lambda(\hat{A}+\hat{B})} \\ &= -e^{-\lambda\hat{B}} \hat{B} e^{-\lambda\hat{A}} e^{\lambda(\hat{A}+\hat{B})} + e^{-\lambda\hat{B}} e^{-\lambda\hat{A}} \hat{B} e^{\lambda(\hat{A}+\hat{B})} \\ &= -e^{-\lambda\hat{B}} e^{-\lambda\hat{A}} \underbrace{e^{\lambda\hat{A}} \hat{B} e^{-\lambda\hat{A}}}_{\text{Weyl form of Hadamard}} e^{\lambda(\hat{A}+\hat{B})} + e^{-\lambda\hat{B}} e^{-\lambda\hat{A}} \hat{B} e^{\lambda(\hat{A}+\hat{B})} \\ &= -\lambda[\hat{A}, \hat{B}] \hat{f}(\lambda). \end{aligned} \quad (8.83)$$

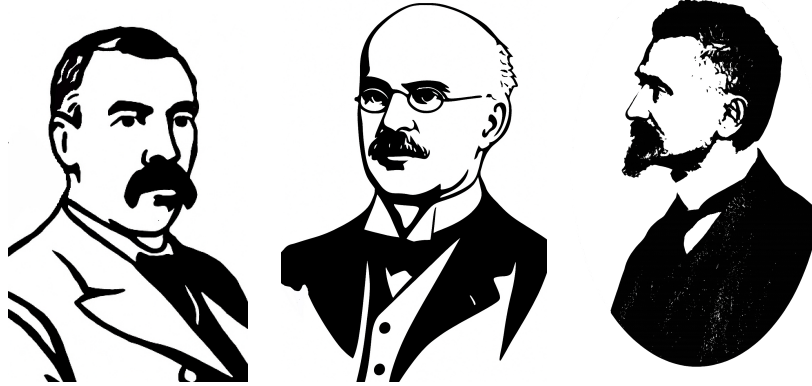


Fig. 8.6: (Left) Henry Baker, (center) John Campbell, and (right) Felix Hausdorff, who discovered the Baker-Campbell-Hausdorff formula. [All images are public domain and have been converted to stencils by the authors.]

Here, for each derivative on the first line, we choose which side to put the operator term on. Then, by using a multiply-by-one, we can form a Hadamard lemma, which is in the Weyl form and can be evaluated. Then finally, we can pull the commutator out to the left because it commutes with $\hat{A}$ and $\hat{B}$. Now, with this result, we can integrate the differential equation via

$$\int d(\ln \hat{f}(\lambda)) = -[\hat{A}, \hat{B}] \int d\lambda \lambda \implies \hat{f}(\lambda) = \hat{f}(\lambda=0) e^{-\frac{1}{2}\lambda^2[\hat{A}, \hat{B}]}. \quad (8.84)$$

Using the fact that the $\hat{f}(0)$ is the identity operator, we can rearrange the results so that we obtain the Weyl form of the Baker-Campbell-Hausdorff formula when we set $\lambda = 1$:

$$e^{\hat{A}+\hat{B}} = e^{\hat{A}} e^{\hat{B}} e^{-\frac{1}{2}[\hat{A}, \hat{B}]}, \text{ when } [\hat{A}, \hat{B}] \text{ commutes with } \hat{A} \text{ and } \hat{B}. \quad (8.85)$$

For us, we pick $\hat{A} = \frac{i}{\hbar} p_0 \hat{x}$, $\hat{B} = -\frac{i}{\hbar} x_0 \hat{p}$, and then $-\frac{1}{2}[\hat{A}, \hat{B}] = -\frac{i x_0 p_0}{2\hbar}$. This then lets us compute the coherent-state wavefunction as

$$\begin{aligned} \langle x | x_0(t), p_0(t) \rangle &= e^{-\frac{i\omega t}{2}} e^{-\frac{i x_0(t) p_0(t)}{2\hbar}} \langle x | e^{\frac{i}{\hbar} p_0(t) \hat{x}} e^{-\frac{i}{\hbar} x_0(t) \hat{p}} | 0 \rangle \\ &= e^{-\frac{i\omega t}{2}} e^{-\frac{i x_0(t) p_0(t)}{2\hbar}} e^{\frac{i}{\hbar} p_0(t) x} \langle x - x_0(t) | 0 \rangle \\ &= \left(\frac{m\omega}{\pi\hbar} \right)^{\frac{1}{4}} e^{-\frac{i\omega t}{2}} e^{-\frac{i x_0(t) p_0(t)}{2\hbar}} e^{\frac{i}{\hbar} p_0(t) x} e^{-\frac{m\omega}{2\hbar} (x - x_0(t))^2}. \end{aligned} \quad (8.86)$$

This wavefunction translates the ground-state wavefunction according to the classical oscillation of the position and it picks up a additional time-dependent complex phases. The complex phases vanish when we look at the probability distribution. So,

this says that the coherent state evolves with the shape of the ground state and with its peak following the classical time evolution of the position. This is why it does not have an uncertainty that changes with time, because it always has the shape of the ground-state wavefunction. So, the average of the coherent state position (and momentum) evolve just as the classical system does, but the quantum system also has uncertainty, which the classical system does not. It is only when we look at the variance of the position and momentum that we find differences between the quantum system and the classical system.

These coherent states are the most natural states to work with in cases where we take a system in its ground state and perturb it, as we can do with an ion in a trap that is hit with a short pulse of light to impulse it. Creating energy eigenstates is often quite difficult to do. It has been done, and even imaged, with trapped atoms [2] for $n = 0, 1$, and 2 , and it is something that can also be done with high precision in ion traps. But, it is much easier to create coherent states in general. This is because the coherent states are the closest states to the classical ones.

8.8 Summary

In this chapter, we have seen how the same mathematical formalism that is used in classical mechanics to decouple the equations of motion of the harmonic oscillator, can be generalized to the quantum regime, where it leads to a natural description of the quantum harmonic oscillator. After developing some useful operator identities, we were able to construct the energy eigenstates and the coherent states—two states that behave quite differently in the quantum realm. The energy eigenstates are so-called stationary states and they do not evolve in time, except for a global time-varying phase, which cannot change any expectation values. The coherent states are closest to the classical system, with the peak of the wavefunction tracing out the classical equation of motion for position, but also satisfying the Heisenberg uncertainty principle, which implies that the position and momentum are indeterminate, with a distinctive variance determined, in part, by $\hbar$. We also learned some new mathematics, including the Rodrigues formula, the Weyl forms of the Hadamard lemma and Baker-Campbell-Hausdorff formula and we learned how to transform from the momentum operator to its matrix representation in terms of derivatives in position space.

Chapter 8 problems

Problem 8.1 Damped harmonic oscillator. A damped harmonic oscillator has the equations of motion given by $\dot{x} = p/m$ and $\dot{p} = -\gamma\dot{x} - m\omega^2x$. Determine the $A(t)$ and $A^*(t)$ solutions and use them to find $x(t)$ and $p(t)$ for the damped harmonic

oscillator. If you are brave, compute the energy $E(t) = \frac{1}{2m}p^2(t) + \frac{1}{2}m\omega^2x^2(t)$ as well.

Problem 8.2 Continuously varying energy. Consider the following superposition of harmonic oscillator energy eigenstates:

$$|\psi(\alpha)\rangle = \alpha|0\rangle + \sqrt{1-|\alpha|^2}|1\rangle. \quad (8.87)$$

Compute the expectation of the energy in the state $|\psi\rangle$ given by $\langle\psi(\alpha)|\hat{H}|\psi(\alpha)\rangle$ for real α with $0 \leq \alpha \leq 1$. Verify that the energy continuously varies with α . Explain how this occurs given that $\hat{H}|0\rangle = \frac{\hbar\omega}{2}|0\rangle$ and $\hat{H}|1\rangle = \frac{3\hbar\omega}{2}|1\rangle$.

Problem 8.3 Momentum matrix of the simple harmonic oscillator. Construct the **P** matrix for the Heisenberg matrix mechanics solution of the simple harmonic oscillator. Once you have the position and momentum matrices, compute $\mathbf{H} = \frac{1}{2m}\mathbf{P}^2 + \frac{1}{2}m\omega^2\mathbf{X}^2$ and verify that it is a diagonal matrix with the diagonal matrix elements being the quantum energy eigenvalues of the simple harmonic oscillator. Finally, compute the canonical commutation relation and verify that it is $i\hbar$ times the identity matrix.

Problem 8.4 Momentum uncertainty of a coherent state. Compute the uncertainty for momentum in the coherent state $|x_0, p_0\rangle$.

Problem 8.5 Uncertainty of simple harmonic oscillator energy eigenstates. Compute the variance in position and in momentum for the energy eigenstates

$$|n\rangle = \frac{1}{\sqrt{(\hbar\omega)^n n!}} \left(\hat{A}^\dagger\right)^n |0\rangle. \quad (8.88)$$

Then compute the uncertainty for each energy eigenstate. Discuss what this means.

Problem 8.6 Classical uncertainty over a period of motion. We can compute uncertainty for the classical harmonic oscillator in time. For simplicity, we will take $p_0 = 0$, so that $x(t) = x_0 \cos(\omega t)$, $p(t) = -x_0 m\omega \sin(\omega t)$, and $E = \frac{1}{2}m\omega^2 x_0^2$. We compute the average position and average time by integrating over a period for $t = 0$ to $t = T = 2\pi/\omega$. You should find both averages vanish. Compute as well the time average of the squares of position and momentum, via

$$\begin{aligned} \langle x^2 \rangle_{\text{cl}} &= \frac{1}{T} \int_0^T dt x_0^2 \cos^2(\omega t) \quad \text{and} \\ \langle p^2 \rangle_{\text{cl}} &= \frac{1}{T} \int_0^T dt m^2 \omega^2 x_0^2 \sin^2(\omega t). \end{aligned} \quad (8.89)$$

Verify that

$$\sqrt{\langle x^2 \rangle_{\text{cl}} \langle p^2 \rangle_{\text{cl}}} = \frac{E}{\omega}. \quad (8.90)$$

Comparing to the previous problem, we can see that the classical uncertainty in time and the quantum uncertainty have the same functional form of the energy divided by the frequency.

Problem 8.7 Position and momentum solutions. Verify the solution to the differential equation for $x(t)$ and $p(t)$ in Eq. (8.60).

Problem 8.8 Average position and momentum of coherent states. In this problem, we compute the time-varying expectation values for the coherent state in a different way. Recalling that

$$x(t) = \langle 0 | \hat{D}^\dagger(x_0, p_0) e^{\frac{i}{\hbar} \hat{H} t} \hat{x} e^{-\frac{i}{\hbar} \hat{H} t} \hat{D}(x_0, p_0) | 0 \rangle, \quad (8.91)$$

compute the average by defining

$$\hat{x}(t) = e^{\frac{i}{\hbar} \hat{H} t} \hat{x} e^{-\frac{i}{\hbar} \hat{H} t}, \quad (8.92)$$

and solving it by differentiating with respect to time, finding the differential equation it solves, and solving it. This is similar, but not identical to what was done in the main text, and it does require you to work with a similarly defined $\hat{p}(t)$ as well. Once you find this operator, evaluating the expectation value should be relatively simple. Also evaluate the same expectation value for the corresponding momentum operator, which you also can immediately solve for.

Problem 8.9 Momentum eigenstates. Prove that the state $|p_0\rangle$ in Eq. (8.66) is an eigenstate of momentum with eigenvalue p_0 .

Problem 8.10 Hermite polynomial. Verify the table of Hermite polynomials in Tab. 8.1 by using the recurrence relations for the Hermite polynomials and the known starting values for H_0 and H_1 .

Problem 8.11 Time evolution of a coherent state. Make an animation of the probability distribution of the coherent state (for some chosen x_0 and p_0) as a function of time t and describe the behavior it has.

Problem 8.12 Deriving the time-independent Schrödinger equation. In this problem, we will use our method to convert the momentum operator into its position-space representation to derive the time-independent Schrödinger equation. We start from the relation that states with definite energy satisfy and then multiply it by a position bra $\langle x|$. Namely, we have

$$\langle x | \frac{\hat{p}^2}{2m} + V(\hat{x}) | \psi_n \rangle = \langle x | E_n | \psi_n \rangle. \quad (8.93)$$

We denote $\langle x | \psi_n \rangle$ as $\psi_n(x)$. Use our identity for the position-space representation of momentum to find a differential equation that $\psi_n(x)$ satisfies in order to be the wavefunction of a state of definite energy. Do not try to solve the equation. This equation is called the time-independent Schrödinger equation.

Problem 8.13 Solving the Schrödinger equation for the harmonic oscillator wavefunctions. In this problem, you will illustrate how to derive the solution of

the simple harmonic oscillator using the Schrödinger equation derived in the previous problem. If you look at the Schrödinger equation for large $|x|$, we can drop the E_n term and it approximately becomes

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi_n(x) \approx \frac{1}{2} m \omega^2 x^2 \psi_n(x), \quad (8.94)$$

which can be solved by a Gaussian function if we consider only the large terms. Verify this.

Next, assume $\psi_n(x) = C_n H_n \left(\sqrt{\frac{m\omega}{\hbar}} x \right) e^{-\frac{m\omega}{2\hbar} x^2}$, where we do not assume that H_n is the Hermite polynomial, just an unknown function here. Find the differential equation for H_n by substituting this form into the Schrödinger equation. Look up the differential equation of the Hermite polynomials to verify that that is what they are in the solution. Note that computing the constant C_n is challenging, but you can find an integral relation of the Hermite polynomials that allows you to do this. It is not needed for this problem though.

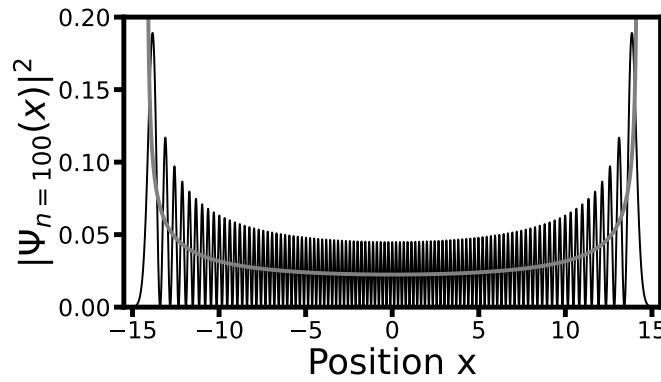


Fig. 8.7: The black solid line plots $|\psi_{n=100}(x)|^2$ (black) and the grey solid line plots the classical probability $1/(\pi\sqrt{x_0^2 - x^2})$ (grey). You are asked to compare the two and see whether you believe the limit as $n \rightarrow \infty$ will appear like a classical system.

Problem 8.14 Classical versus quantum probability for the simple harmonic oscillator. Consider Fig. 8.7, which plots the probability distribution of an $n = 100$ energy eigenstate of the simple harmonic oscillator along with the classical probability worked out in the main text. Write an essay discussing whether you believe that in the large n limit the energy eigenstates approach the classical result, or do not.

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